MIL-E-1/193C
23 June 1977
SUPERSEDING
MIL-E-1/193B
10 April 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

© TYPES OC3A AND OC3W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage regulator

Outline
OC3A --- 9-7 (EIA)
OC3W --- See figure 1

Base
OC3A --- B6-8
OC3W --- See figure 1

Envelope --- T9

Cathode --- Glow discharge

Base connections:

Pin No.	---	1	2	3	5	7	8
Element	---	nc	k	7 (Note 1)	a	3 (Note 1)	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ebb	Starting voltage	Operating voltage	Operating current	Alt
Unit:	Vdc	Vdc	Vdc	mAdc	ft
Maximum:	---	---	105 (approx)	40	10,000
Minimum:	---	133	---	5	---
TEST CONDITIONS:	300	---	---	--- (Note 2)	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3347	Ionization voltage (1) (with illumination)	3, 4	Rp/Ib = 5 to 40 mAdc	0.65	II	Ez	---	127	Vdc
3337	Voltage drop (1)	-	Rp/Ib = 40 mAdc	0.65	II	Eb	---	112	Vdc
3337	Voltage drop (3)	-	Rp/Ib = 5 mAdc	0.65	II	Eb	105	---	Vdc
3335	Regulation (1)	-	Eb(1)-Eb(3)	0.65	II	---	---	4.0	Vdc
3278	Noise and oscillation (oscillation test)		Esig = 100 mVac; Rl = 500 ohms; Ebb/Ib = 5 to 40 mAdc	0.65	II	---	---	---	---
3278	Noise and oscillation (noise test)	-	Ebb/Ib = 40 mAdc; Rl = 500 ohms	0.65	II	ΔEb	---	15.0	mVac
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1041	Shock (Type OC3W)	-	900 G	15	S2	---	---	---	---
	Shock-test end point (Type OC3W):								
3347	Ionization voltage (1)	-		---	---	Ez	---	133	Vdc
3337	Voltage drop (2)	-		---	---	Eb	---	113	Vdc
3337	Voltage drop (3)	-		---	---	Eb	103	---	Vdc
3335	Regulation (1)	-		---	---	---	---	4.0	Vdc
3335	Regulation (2)	-		---	---	---	---	2.5	Vdc
3337	Voltage drop (2)	-	Rp/Ib = 30 mAdc	---	---	Eb	---	111	Vdc
3335	Regulation (2)	-	Eb(2)-Eb(3)	---	---	---	---	2.0	Vdc
3305	Leakage current	-	Eb = 50 Vdc	---	---	Ilb	---	10.0	μAdc
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	5		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group A; Rp/Ib = 30 mAdc	---	---	---	---	---	---
---	Life-test end points (500 hours):								
3347	Ionization voltage (1)	-		---	---	Ez	---	133	Vdc
3337	Voltage drop (2)	-		---	---	Eb	---	113	Vdc
3337	Voltage drop (3)	-		---	---	Eb	103	---	Vdc
3335	Regulation (1)	6		---	---	---	---	4.0	Vdc
3335	Regulation (2)	-		---	---	---	---	2.5	Vdc
1031	Variable-frequency vibration (Type OC3W)	7	No voltages applied	---	---	---	---	---	---
3347	Ionization voltage (2) (total darkness)	7, 8	Rp/Ib = 5 to 40 mAdc	---	---	Ez	---	210	Vdc

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NOTES:

1. Pins 3 and 7 tied internally.
2. Fixed resistors may be used and Ebb varied to give desired current.
3. Test to be conducted with the tube exposed to light intensity of 5 to 50 footcandles.
4. This test to be performed at the conclusion of the holding period.
5. This test applies for flexible leads as well as for rigid leads.
6. The tube shall have reached the end of its life with regard to regulation (1) when that regulation first exceeds the value of 4.0 volts.
7. (C) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
8. (C) Test to be conducted with the tube under test enclosed in a light tight enclosure composed of a nonconducting material. Measurements to be performed after the test specimen is held in the enclosure for 24 hours. The tube shall fire within 15 seconds.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
Army - MI, MU
DLA - ES

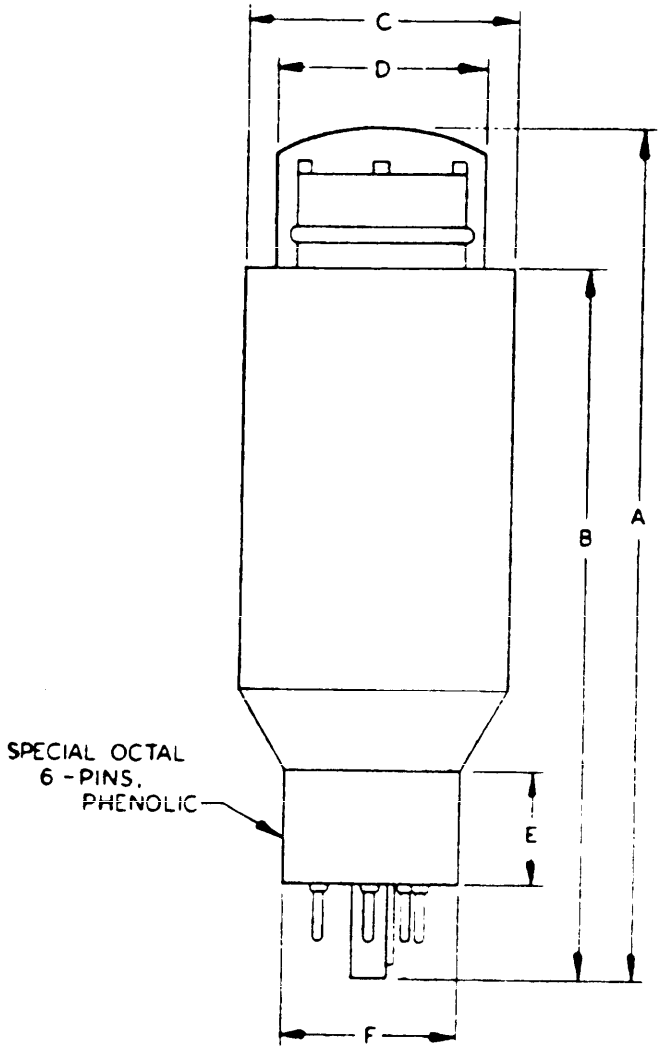
User activities:
Navy - AS, OS, MC, CG
Air Force - 11, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5960-3143)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 3 (SEE NOTE 7)		
A	---	4.13 (104.90)
B	---	3.403 (86.44)
C	---	1.57 (39.88)
D	---	1.32 (33.53)
E	.843 (21.41) Nom	
F	1.136 (28.85)	1.176 (29.87)

FIGURE 1. Outline drawing of electron tube type OC3W.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/193C ELECTRON TUBE, RECEIVING TYPES 0C3A AND 0C3W

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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